Notes: Hortaçsu, Kastl, and Zhang (AER, 2018)

Bid Shading and Bidder Surplus in the US Treasury Auction System

Contents

1 Big picture

- **Question.** Why do primary dealers bid systematically *higher yields* (lower prices) than other bidders in US Treasury uniform-price auctions? Is that because they value the securities less, or because they can exercise more market power through bid shading?
- **Setting.** The paper studies individual bid-level data from US Treasury bill and note auctions between July 2009 and October 2013. The auctions are sealed-bid, uniform-price share auctions in which bidders submit step-function demand schedules.
- **Core challenge.** A bidder's observed bid mixes together two things: true willingness to pay and strategic shading. In this market the problem is especially subtle because bids are *multi-unit* demand schedules, not single bids, and primary dealers have an informational advantage because they observe customer (indirect) bids routed through them.
- **Main method.** The paper adapts the Wilson (1979) share-auction model, with Kastl's (2011) discrete-bid extension, and further incorporates the information asymmetry between primary dealers, direct bidders, and indirect bidders. The key empirical object is the *expected residual supply elasticity* faced by each bidder.
- **Main descriptive fact.** Primary dealers bid higher yields than both direct and indirect bidders across essentially all maturities. Raw bid differences range from roughly 2 to 11 basis points depending on the security.
- **Main structural result.** Primary dealers do *not* appear to bid higher yields because they value the securities less. In fact, after controlling for bidder size, they tend to have *higher* willingness to pay than other bidders, but their ability to shade bids is even larger, especially for longer-maturity notes.
- **Main surplus result.** Total bidder surplus averages about **3 basis points** of issue size over the sample, with much larger surpluses in Treasury notes than in Treasury bills.
- **Main efficiency result.** Allocative inefficiency is fairly modest on average, about **2 basis points**, though it is much larger for long-maturity notes than for short-maturity bills.
- **Economic takeaway.** Even under a uniform-price auction, large and informed bidders can exercise meaningful market power. The Treasury auction system is fairly efficient, but primary dealers retain substantial informational and strategic advantages.

2 Data and measurement (key objects)

2.1 Institutional setting and sample

- **Auction format.** The Treasury uses sealed-bid *uniform-price* auctions. Competitive bidders submit price-quantity schedules. Noncompetitive bids are also accepted, but are capped at $5 million and are due earlier than competitive bids.
- **Why uniform-price matters.** The Treasury switched from discriminatory/pay-as-bid auctions to the uniform-price format in 1998 after experimentation with two- and five-year notes. The hope was to reduce bid-shading incentives and improve participation.
- **Sample.** The paper uses **975 auctions** between July 2009 and October 2013:

  - 822 auctions of cash-management bills and standard bills,
  - 153 auctions of 2-year, 5-year, and 10-year notes.

- **Issue volume.** Total issuance in the sample is about **$27.3 trillion**, with average issue size around $28 billion.
- **Maturities covered.** The sample includes CMBs, 4-week, 13-week, 26-week, and 52-week bills, plus 2-year, 5-year, and 10-year notes.

2.2 Bidder classes and information structure

- **Primary dealers (PDs).** During the sample, 17 to 21 primary dealers regularly bid in Treasury auctions and make markets in Treasuries.
- **Direct bidders (DBs).** Non-primary dealers who submit bids directly for their own account. This category includes brokers, dealers, mutual funds, pension funds, insurance companies, depository institutions, foreign entities, the Federal Reserve's SOMA account, and individuals.
- **Indirect bidders (IBs).** Investors who submit bids through a primary dealer. These include large asset managers, pension funds, and other institutional investors.
- **Key asymmetry.** PDs not only bid for their own house accounts, they also route indirect bids. Because they observe those customer bids before submitting their own bids, they have more precise information about residual supply than other bidders.
- **Bid format.** Bids are step functions with minimum yield increments of 0.5 basis points for 13-, 26-, and 52-week bills and cash-management bills, and 0.1 basis points for the other maturities.
- **Allocation rule.** The paper assumes the practical Treasury rule of *pro-rata on-the-margin rationing* when demand exceeds supply at the clearing price. In the extremely rare event of multiple clearing prices due to quantity discreteness, the highest clearing price is selected.

2.3 Descriptive objects from the bids

- **Quantity-weighted bid yield.** For each bidder, the paper summarizes the bid schedule with a quantity-weighted average bid yield.
- **Within-auction bid dispersion.** The within-auction standard deviation of quantity-weighted bid yields is used to summarize how dispersed bids are within a bidder group.
- **Percent of issue size.** The total quantity tendered by a bidder as a share of auction size is a key measure of bidder size.
- **Percent of tender won.** The share of a bidder's tender quantity that is actually allocated in the auction.
- **Normalized bid.** Quantity-weighted bid yield minus the Treasury's published yield for that maturity on the auction day. This compares bidding to prevailing market rates.

2.4 Key descriptive facts

- **Tender shares (Table 1).** Primary dealers submit most of the demand:

- about **69% to 88%** of tendered quantities,
- direct bidders submit about **6% to 13%**,
- indirect bidders submit about **6% to 18%**.

- **Allocation shares (Table 1).** PDs win a smaller share of their tender than other bidders:
  - PDs are allocated **46% to 76%** of competitive demand,
  - DBs roughly **7% to 17%**,
  - IBs a disproportionate **17% to 40%** of winning quantities relative to their tender shares.
- **Raw bid levels (Table 2).** PDs systematically bid higher yields than both DBs and IBs across all maturities. The PD–IB yield gap ranges from roughly **3 to 18 basis points**; the PD–DB gap ranges from roughly **2 to 8 basis points**, except for 2-year notes.
- **Demand sizes (Table 2).** PDs typically bid for much larger quantities:
  - PDs demand about **10% to 20%** of issue size,
  - DBs about **3% to 5%**,
  - IBs about **1% to 3%**.
- **Tender success (Table 2).** PDs win only about **20%** of their tendered quantities, compared with roughly **40%–50%** for direct bidders and **70%–80%** for indirect bidders.
- **Normalized bids (Table 3).** Relative to prevailing Treasury yields on the day of auction:
  - indirect bidders tend to bid *below* the market yield,
  - direct bidders bid around the market yield,
  - primary dealers bid *above* the market yield.

2.5 Unobservables and economic interpretation

- **Why valuations differ.** Treasury securities may have different values across bidders because of buy-and-hold motives, repo collateral value, dealer inventory needs, complementary value from primary-dealer status, or different portfolio needs.
- **Private versus common values.** The paper argues that a private-values interpretation is plausible because bidders have access to the same macro and market data and there is an active when-issued market that already aggregates much common information.
- **Remaining concern.** A formal test of private values versus common values would be desirable, but the US data do not contain the bid revisions needed to implement the type of test the authors used in earlier Canadian Treasury work.
- **Auction-specific heterogeneity.** Demand conditions can vary substantially across auctions, especially in 2009 and 2010, so the authors estimate marginal valuations auction by auction rather than pool auctions of the same maturity.

3 Models and empirical specifications (all equations + interpretation)

3.1 Share-auction environment

- There are three bidder classes: primary dealers, direct bidders, and indirect bidders.
- Each bidder submits a *demand schedule* for a perfectly divisible security with random supply Q.
- Indirect bidders submit bids first to their primary dealer. Direct bidders and primary dealers then submit bids to the Treasury, with primary dealers conditioning on the indirect orders they observe.
- A bidder of type θ_i who wins quantity Q_i^c at clearing price P^c has expected utility

$$EU_i(\theta_i) = \mathbb{E}\left[\int_0^{Q_i^c} v_i(u, \theta_i)\, du - P^c Q_i^c \mid \theta_i\right].$$

- *Interpretation.* Treasury auctions are not one-unit auctions. Bidders care about the whole inframarginal area under their demand curves, not just whether they win or lose one object.

3.2 Distribution of the market-clearing price

- The strategic object faced by a bidder is the **distribution of the market-clearing price** conditional on her information.
- For a direct bidder, the probability that the clearing price is below some price p is the probability that residual supply at price p exceeds her own demanded quantity at that price.
- For a primary dealer, this distribution conditions on the customer bids the dealer observes. That is the source of the dealer's informational advantage.
- *Interpretation.* The paper does not directly observe a bidder's demand curve, but it can estimate the residual supply the bidder expects to face by resampling rival bids in the data.

3.3 Optimal bidding condition

The necessary condition for optimal bidding in the discrete share-auction model is

$$v_i(q_k, \theta_i) = E(P^c \mid b_k > P^c > b_{k+1}, \theta_i) + \frac{q_k}{\Pr(b_k > P^c > b_{k+1} \mid \theta_i)} \frac{\partial E(P^c;\, b_k \geq P^c \geq b_{k+1} \mid \theta_i)}{\partial q_k}.$$

- The first term is the **expected market-clearing price** conditional on the kth bid step being marginal.
- The second term is the **markup / shading term**. It depends on the slope of expected residual supply, and so resembles the inverse-elasticity markup rule in monopsony or oligopoly.
- If the bidder is effectively a price taker, the residual supply is flat, the second term is close to zero, and the bidder should bid close to true value.

- If the bidder faces a steep residual supply curve, shading is profitable because reducing demanded quantity pushes down the clearing price on all inframarginal units.
- *Interpretation.* This is the paper's key decomposition result: a bid can be low either because value is low or because the bidder shades heavily. The residual supply elasticity tells us which one is at work.

3.4 Two definitions of bid shading

The paper defines two related shading measures.

Definition 1: Average bid shading

$$B(\theta_i) = \frac{\sum_{k=1}^{K_i} q_k \left[v_i(q_k, \theta_i) - b_k \right]}{\sum_{k=1}^{K_i} q_k}.$$

- This is the quantity-weighted difference between marginal value and the actual bid.
- It can be negative because of bid discreteness: a bidder may submit a step whose bid lies above marginal value even though overall surplus remains positive.

Definition 2: Average shading factor

$$S(\theta_i) = \frac{\sum_{k=1}^{K_i} q_k \left[v_i(q_k, \theta_i) - E(P^c \mid b_k > P^c > b_{k+1}, \theta_i) \right]}{\sum_{k=1}^{K_i} q_k}.$$

- This compares marginal value with the expected clearing price when the kth step is marginal.
- It is nonnegative and has a clean interpretation as the expected inverse elasticity of residual supply faced by the bidder.

3.5 Estimating marginal valuations

- The paper uses a **resampling** method developed in earlier auction work by Hortaçsu, Kastl, and coauthors.
- For direct and indirect bidders, the authors repeatedly draw rival bids from the empirical distribution of bids within an auction to simulate the residual supply curve and thus the clearing-price distribution.
- For primary dealers, the procedure is modified because PD bids depend on observed indirect bids. The authors kernel-weight the resampling so that a dealer is compared to other dealers who observed similar customer order flow.
- The slope term in the optimality condition is estimated using a numerical derivative of the expected clearing-price object.
- Estimation is done **auction by auction**, not pooled across many auctions. This is designed to reduce bias from auction-level unobservables, changing demand conditions, or multiple strategic equilibria across auctions.

3.6 Surplus and efficiency calculations

- Once a bidder's rationalizing marginal values are estimated, the paper computes bidder surplus as the area under the marginal value curve up to the allocated quantity, minus the payment made:
$$\text{Surplus}_i = \int_0^{Q_i^c} \bar{v}_i(u)\,du - P^c Q_i^c,$$
where $\bar{v}_i$ is the upper envelope of the estimated stepwise marginal valuation function.
- Allocative efficiency is computed by comparing the realized allocation to the allocation that would maximize total surplus given estimated marginal values:
$$\text{Efficiency loss} = \text{efficient surplus} - \text{actual surplus}.$$
- *Important interpretation.* Bidder surplus is not itself "waste." Any voluntary mechanism must leave bidders some inframarginal surplus, especially with downward-sloping demand. So the observed surplus is a *conservative upper bound* on how much the Treasury could save by improving the mechanism.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies market power versus willingness to pay

- Raw bids alone do not separate market power from demand. A bidder can bid a high yield either because value is low or because shading is strong.
- The key identifying object is the **expected residual supply elasticity** around the bidder's submitted bid.
- This object is driven by the *shapes of other bidders' bids*, which are directly observed in the data.
- If all rival bids were nearly identical and the auction oversubscribed, each bidder would effectively be a price taker and shading would be minimal.
- Therefore, heterogeneity in observed rival bid schedules identifies how much potential market power a bidder has.

4.2 Why primary dealers can look different

- PDs are larger bidders on average, and larger bidders can face less elastic residual supply and therefore have greater incentive to shade.
- PDs also see indirect bids submitted through them, so their beliefs about residual supply are more precise than those of other participants.
- This information edge helps explain why PDs may bid more strategically even if their underlying valuations are high.

4.3 Why the private-values interpretation is plausible

- The paper assumes a private-values environment rather than a common-values one.
- The argument is that Treasury auction bidders all observe the same public macro and market data, and the active when-issued market aggregates much dispersed information before the auction.
- Thus, the remaining heterogeneity at the auction stage is plausibly due to inventory needs, collateral motives, or portfolio-specific demand rather than disagreement about one latent common value.

4.4 Main assumptions and caveats

- **Optimization.** Bidders are assumed to choose bids to maximize expected surplus.
- **Bayesian-Nash equilibrium.** The realized distribution of rival bids is used as an empirical proxy for ex ante beliefs about residual supply.
- **Unilateral noncooperative behavior.** The estimated shading magnitudes should be interpreted under the assumption that bidders individually exploit available market power.
- **No direct common-value test.** The paper cannot test private versus common values in the US data because it does not observe bid revisions.
- **Possible information sharing among PDs.** The paper notes recent allegations that dealers may have shared order-flow information, but this is not directly modeled in the baseline analysis.
- **Precision versus robustness trade-off.** Auction-by-auction estimation reduces precision, but it is more robust to unobserved heterogeneity and equilibrium differences across auctions.

4.5 Relation to the literature

- **Auction theory.** The paper builds on Wilson (1979), Friedman (1960), Vickrey (1961), and Kastl (2011), translating multi-unit auction theory into an empirically implementable framework.
- **Treasury auction policy.** It differs from the Treasury's own empirical assessments, such as Malvey and Archibald (1998), by focusing on *inframarginal bidder surplus* rather than when-issued prices.
- **Empirical auctions.** Methodologically, it uses the GPV-style insight that equilibrium behavior can reveal latent values, but adapts it to multi-unit, discrete-bid, information-asymmetric Treasury auctions.

5 Tables and figures

5.1 Figure 1: Intuition for bid shading

Read:

- Figure 1 is the paper's core economic picture. A bidder with downward-sloping demand facing an upward-sloping residual supply curve chooses a bid point (P^*, Q^*) below the competitive point (P^{comp}, Q^{comp}).
- The shaded area under the demand curve and above the bid point is bidder surplus.
- The steeper the residual supply curve, the greater the incentive to reduce demanded quantity and push down the market-clearing price.
- This figure makes clear that bid shading is not about "mistakes" but about strategic quantity reduction when the bidder can influence price.

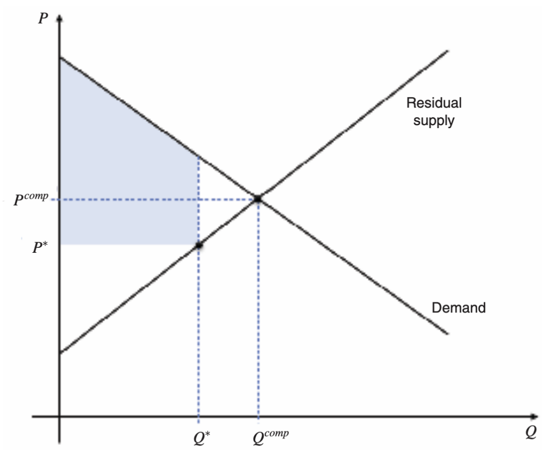

Figure 1. Illustration of Bid Shading when Residual Supply Is Known

Table 1—Summary Statistics

Maturity	No. auctions	Mean issue size (billion)	Percent demand tendered: Primary	Direct	Indirect	Percent won: Primary	Direct	Indirect
CMBs	101	24.6	88	6	6	76	7	17
4 week	222	31.9	85	7	8	65	8	27
13 week	222	28.5	86	7	7	67	8	25
26 week	222	24.9	84	7	9	61	8	31
52 week	55	24.3	82	8	10	61	10	29
2 year	51	34.7	73	13	14	55	17	28
5 year	51	34.3	70	12	18	48	12	40
10 year	51	20.5	69	13	18	46	16	38

Notes: "CMBs" refers to cash management bills. "Primary" refers to primary dealers; "direct" to direct bidders; and "indirect" to indirect bidders.

5.2 Table 1: Summary statistics

Read:

- Table 1 gives the market structure of the auctions. PDs dominate total demand, tendering 69% to 88% of quantities depending on maturity.
- Yet PDs win a smaller share of what they ask for than indirect bidders. That is already a strong sign that PDs bid less aggressively on price.
- For example, in 10-year notes PDs tender 69% of demand but win only 46%, whereas indirect bidders tender 18% and win 38%.
- This contrast motivates the entire paper: are PDs simply less interested, or are they strategically shading more?

5.3 Tables 2 and 3: Description of bids and normalized bids

Read:

- Table 2 shows the raw bid facts. PDs bid higher yields than everyone else, with especially large gaps relative to indirect bidders.

Table 2—Description of Bids

	Bid			Within auction SD[Bid]			Percent of issue size			Percent of tender won		
Maturity	Primary	Direct	Indirect	Primary	Direct	Indirect	Primary	Direct	Indirect	Primary	Direct	Indirect
CMBs	0.1501	0.1389	0.1185	0.0244	0.0201	0.0223	19	5	3	21	36	64
4 week	0.0943	0.0699	0.0463	0.0254	0.0337	0.0266	18	3	2	19	52	84
13 week	0.1119	0.0866	0.0683	0.0248	0.0332	0.0249	19	3	2	19	54	84
26 week	0.165	0.1368	0.1254	0.0275	0.0391	0.0272	20	4	2	16	52	71
52 week	0.2617	0.2356	0.227	0.0299	0.0333	0.017	17	4	2	20	47	67
2 year	0.5604	0.5231	0.4927	0.0397	0.046	0.0939	13	4	1	22	42	70
5 year	1.5627	1.4902	1.4384	0.0682	0.0631	0.1244	10	3	1	24	55	82
10 year	2.7229	2.6482	2.5906	0.0732	0.0706	0.192	11	3	1	21	50	71

Notes: "Bid" refers to quantity-weighted bid-yield averaged across auctions. "Within-auction SD[Bid]" refers to the within-auction standard deviation of quantity-weighted bid-yield, averaged across auctions. "Percent of issue size" refers to the percentage of total issue size demanded by a bidder, averaged across auctions. "Percent of tender won" refers to the percentage of the tender quantity that the bidder won at the auction, averaged across auctions. "CMBs" refers to cash management bills. "Primary" refers to primary dealers; "direct" to direct bidders; and "indirect" to indirect bidders.

Table 3—Description of Normalized Bids

	Normalized bid			Within auction SD[norm bid]		
Maturity	Primary	Direct	Indirect	Primary	Direct	Indirect
4 week	2.04	−0.36	−2.73	2.53	3.37	2.66
13 week	1.99	−0.53	−2.36	2.48	3.32	2.49
26 week	2.33	−0.48	−16.3	2.75	3.91	2.72
52 week	2.68	0.07	−0.79	2.99	3.33	1.70
2 year	4.84	11.12	−2.36	3.97	4.60	9.39
5 year	8.01	0.77	−4.41	6.82	6.31	12.44
10 year	7.84	0.37	−5.39	7.32	7.06	19.20

Notes: Normalized bids are defined as quantity-weighted bids (in basis points) minus the interest rate of the corresponding maturity reported by the US Treasury on the day of the auction. "Normalized bid" refers to quantity-weighted normalized bid-yield averaged across auctions. "Within auction SD[norm bid]" refers to the within-auction standard deviation of quantity-weighted normalized bid-yield, averaged across auctions. "Primary" refers to primary dealers; "direct" to direct bidders; and "indirect" to indirect bidders.

- Indirect bids are more dispersed than PD bids, especially for long maturities. For 10-year notes, the within-auction standard deviation of indirect bids reaches 19.20 basis points.
- PDs bid for much larger quantities—10% to 20% of issue size—whereas DBs and IBs demand only a few percent.
- Table 3 normalizes bids by prevailing Treasury yields. It shows that IBs typically bid below market yields, DBs near market yields, and PDs above market yields. This strengthens the interpretation that PDs are the least aggressive on price.

5.4 Table 4: Analysis of bid differentials

Read:

- Table 4 formalizes the descriptive bid gaps with auction fixed effects.
- In bills, PDs bid about **2.46 basis points** higher than direct bidders and **4.20 basis points** higher than indirect bidders. In notes, the corresponding raw differences are **5.97** and **10.89** basis points.
- Adding bidder size shrinks the differences but does not eliminate them. The residual gaps are about **0.93** and **2.53** basis points in bills, and **0.97** and **4.44** basis points in notes.
- Bidder size itself matters a lot. The paper interprets the coefficients as showing that increasing market share from 0 to 10% of issue size is associated with about **1 basis point** higher bid yield in bills and **6 basis points** higher bid yield in notes.
- But this by itself is not evidence of market power, because larger bidders may also simply have different demand. The structural decomposition is needed.

Table 4—Analysis of Bids

	Bills		Notes	
	QwBid(bp)	QwBid(bp)	QwBid(bp)	QwBid(bp)
	(1)	(2)	(3)	(4)
Direct	−2.457	−0.929	−5.974	−0.965
	(0.0580)	(0.0600)	(0.270)	(0.314)
Indirect	−4.204	−2.529	−10.89	−4.437
	(0.0604)	(0.0613)	(0.356)	(0.399)
%Q Total		10.04		61.75
		(0.219)		(5.452)
Constant	13.87	11.99	172.0	165.0
	(0.0316)	(0.0426)	(0.261)	(0.460)
Observations	41,359	41,359	13,692	13,692
R^2	0.254	0.289	0.086	0.099
Number of auctions	822	822	153	153

Notes: "QwBid(bp)" refers to the quantity-weighted bids, reported in basis points. %Q Total is the total (maximal) demand. Auction fixed effects are controlled for in every specification. Robust standard errors, clustered by auctions, are reported in the parentheses.

Table 5—Analysis of Bid Shading

	Bills				Notes			
	Shade 1		Shade 2		Shade 1		Shade 2	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Direct	−1.904	−1.434	−0.862	−0.771	−4.696	−2.203	−0.0954	−0.0480
	(0.0851)	(0.0969)	(0.0727)	(0.0884)	(0.305)	(0.255)	(0.0103)	(0.0105)
Indirect	−3.511	−2.996	−1.125	−1.025	−13.36	−10.15	−0.122	−0.0608
	(0.105)	(0.117)	(0.0813)	(0.0978)	(0.684)	(0.469)	(0.0116)	(0.0129)
%Q Total		3.085		0.600		30.73		0.584
		(0.353)		(0.330)		(4.399)		(0.108)
Constant	0.841	0.265	1.174	1.062	−1.888	−5.394	0.125	0.0579
	(0.0543)	(0.0819)	(0.0441)	(0.0756)	(0.478)	(0.353)	(0.00883)	(0.0122)
Observations	41,264	41,264	41,264	41,264	13,692	13,692	13,692	13,692
R^2	0.095	0.097	0.015	0.015	0.158	0.162	0.062	0.069
Number of auctions	822	822	822	822	153	153	153	153

Notes: %Q Total is the total (maximal) demand. Shade 1 ($\equiv B(\theta)$) is as in Definition 1 and Shade 2 ($\equiv S(\theta)$) is as in Definition 2. Bid shading is reported in basis points in columns 1 and 2. Auction fixed effects are controlled for in every specification. Robust standard errors, clustered by auctions, are reported in the parentheses.

Table 6—Analysis of Bid Shading at the Marginal Step

	Bills				Notes			
	Shade 1		Shade 2		Shade 1		Shade 2	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Direct	−1.237	−1.085	−1.940	−1.841	−2.428	−2.345	−0.0185	−0.0141
	(0.0459)	(0.0446)	(0.577)	(0.570)	(0.167)	(0.165)	(0.0019)	(0.0015)
Indirect	−2.192	−2.049	−2.619	−2.526	−5.844	−5.661	−0.022	−0.012
	(0.0497)	(0.0456)	(0.589)	(0.584)	(0.217)	(0.191)	(0.002)	(0.0013)
%Q Marginal		6.214		4.041		8.960		0.475
		(0.554)		(4.333)		(2.468)		(0.0659)
Constant	−0.217	−0.474	2.693	2.526	−0.241	−0.485	0.0218	0.0089
	(0.0225)	(0.0240)	(0.312)	(0.336)	(0.136)	(0.110)	(0.00146)	(0.00113)
Observations	36,860	36,860	36,860	36,860	12,034	12,034	12,034	12,034
R^2	0.161	0.169	0.002	0.002	0.047	0.047	0.044	0.088
Number of auctions	822	822	822	822	153	153	153	153

Notes: %Q Total is the total (maximal) demand. Shade 1 ($\equiv B(\theta)$) is as in Definition 1 and Shade 2 ($\equiv S(\theta)$) is as in Definition 2. Bid shading is reported in basis points in columns 1 and 2. Auction fixed effects are controlled for in every specification. Robust standard errors, clustered by auctions, are reported in the parentheses.

5.5 Tables 5 and 6: Bid shading estimates

Read:

- Table 5 is the key structural evidence. It shows that PDs shade more than DBs and IBs.
- In bills, controlling for size, PDs shade about **1.4 basis points** more than direct bidders and about **3 basis points** more than indirect bidders.
- In notes, the shading differences are much larger: roughly **2 basis points** relative to direct bidders and **10 basis points** relative to indirect bidders after controlling for size.
- Larger bidders shade more. Going from 0 to 10% market share raises shading by about **0.3 basis points** in bills and about **3 basis points** in notes.
- Table 6 shows that the same ranking holds when shading is measured only at the *marginal step*. So the result is not driven by weird out-of-the-money bid steps.
- The paper's decomposition then implies a striking result: PDs actually have *higher* willingness to pay than other bidders, but shade enough to end up submitting higher-yield bids. Controlling for size, the implied PD willingness to pay is about **0.4 (vs. DB) and 0.5 (vs. IB) basis points higher** for bills, and about **1 and 6 basis points higher** for notes.

5.6 Tables 7 and 8: Bidder surplus

TABLE 7—BIDDER SURPLUSES: JULY 2009–OCTOBER 2013

	Primary		Direct		Indirect	
	(bp)	(M$)	(bp)	(M$)	(bp)	(M$)
CMBs	0.17	40.6	0.02	3.8	0.04	9.6
4 week	0.04	26.2	0.00	2.1	0.002	1.1
13 week	0.13	86.4	0.02	11.1	0.008	5.3
26 week	0.33	183	0.03	15.1	0.026	14.6
52 week	0.68	90.8	0.08	10.5	0.14	18.4
2 year	7.40	1,310	1.15	202	0.91	161
5 year	13.07	2,280	1.87	326	1.39	243
10 year	22.22	2,320	3.58	373	1.73	180
Overall	2.3	6,337	0.35	943.5	0.23	633

Notes: "CMBs" refers to the cash management bills, "(bp)" refers to the basis points, "(M$)" to the million US dollars. "Primary" refers to the primary dealers, "direct" to the direct bidders, "indirect" to the indirect bidders.

TABLE 8—ANALYSIS OF BIDDER SURPLUSES

	Bills: Surplus			Notes: Surplus		
	(1)	(2)	(3)	(4)	(5)	(6)
Direct	−18.67	−12.14	−10.61	−1,466	−792.6	−443.4
	(1.662)	(1.725)	(1.566)	(126.9)	(89.68)	(74.97)
Indirect	−25.08	−17.92	−15.61	−1,970	−1,101	−697.9
	(2.424)	(2.403)	(2.034)	(162.3)	(115.6)	(97.91)
%Q Total		42.90	20.29		8,305	7,087
		(8.635)	(9.028)		(1,112)	(1,024)
Number indirects observed			6.615			201.4
			(1.609)			(27.60)
Constant	25.73	17.71	16.07	2,007	1,059	669.9
	(1.237)	(1.678)	(1.586)	(121.6)	(96.24)	(95.08)
Observations	41,264	41,264	41,264	13,692	13,692	13,692
R^2	0.032	0.034	0.041	0.329	0.359	0.392
Number of auctions	822	822	822	153	153	153

Notes: Surpluses are in $1,000s. %Q Total is the total (maximal) demand. Auction fixed effects are controlled for in every specification. Robust standard errors, clustered by auctions, are reported in the parentheses.

Read:

- Table 7 shows that bidder surplus is tiny for bills and much larger for notes.
- Overall surplus over the sample is:
  - **PDs:** 2.3 basis points, or about **$6.34 billion**;
  - **DBs:** 0.35 basis points, or about **$944 million**;
  - **IBs:** 0.23 basis points, or about **$633 million**.
- For PDs, note-auction surplus is especially large: **7.40 bp** in 2-year notes, **13.07 bp** in 5-year notes, and **22.22 bp** in 10-year notes.
- Table 8 shows that PDs' surplus advantage survives controls for bidder size.
- In notes auctions, moving from 0 to 10% of issue size is associated with about **$830,000** more surplus.
- Observing additional indirect bidders is also valuable to PDs. In notes, each additional indirect bidder routed through a PD is associated with about **$200,000** more surplus. Since PDs observe

TABLE 9—ALLOCATIVE EFFICIENCY OF THE AUCTIONS

Maturity	Efficiency loss (in bp)
1 month	0.67
3 months	0.68
6 months	0.76
12 months	0.65
2 year	2.08
5 year	4.50
10 year	6.41

Note: Efficiency loss is the amount of surplus lost due to misallocating the notes/bills.

about 2.5 indirect bidders on average in notes, the paper interprets this as roughly **$500,000**, or about **25%** of PD surplus in notes, plausibly linked to order-flow information—with caveats.

5.7 Table 9: Allocative efficiency

Read:

- Efficiency losses are modest overall, just above **2 basis points** on average.
- The short end of the maturity spectrum is very efficient:
  - 1-month bills: **0.67 bp**,
  - 3-month bills: **0.68 bp**,
  - 6-month bills: **0.76 bp**,
  - 12-month bills: **0.65 bp**.
- Efficiency loss rises sharply with maturity:
  - 2-year notes: **2.08 bp**,
  - 5-year notes: **4.50 bp**,
  - 10-year notes: **6.41 bp**.
- The main interpretation is that longer-maturity securities have more heterogeneous demand and more dispersed bids, so strategic distortions matter more there.

6 Short summary

- **Method.** The paper estimates a structural multi-unit auction model of Treasury bidding with discrete demand schedules and asymmetric information for primary dealers.
- **Identification.** The decomposition uses the expected residual supply elasticity implied by rival bids to separate willingness to pay from bid shading.
- **Main empirical finding.** Primary dealers bid higher yields than direct and indirect bidders not because they value Treasuries less, but because they can shade their bids more while still having relatively high willingness to pay.
- **Welfare finding.** Total bidder surplus is around 3 basis points and efficiency losses are around 2 basis points on average, implying that about 5 basis points is a conservative upper bound on how much the Treasury could save from a better mechanism.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that meaningful bidder market power remains present even in the modern US Treasury *uniform-price* auction system.
- It also shows that the observed high-yield bidding of primary dealers should not be read as evidence that they value securities less. Once demand and strategic shading are separated, PDs often appear to have higher values but stronger strategic incentives.
- Finally, the paper provides a quantitative benchmark for Treasury auction design by measuring bidder surplus and allocative inefficiency directly from a structural model.

7.2 Economic intuition

- A large bidder with downward-sloping demand cares not just about whether it wins, but also about the price paid on all inframarginal units.
- If reducing demanded quantity moves the market-clearing price down, the bidder has an incentive to understate demand. That is bid shading.
- Primary dealers are the natural bidders for whom this force is strongest because they bid for larger quantities and observe extra information about indirect order flow.
- The result is especially strong for notes, where valuations and uses of the securities are more heterogeneous than for bills.

7.3 Takeaway

This paper connects a classic auction-theory idea—residual supply elasticity and monopsony-style bid shading—to one of the largest real-world financial markets. The main message is subtle but important: the Treasury's auction system is fairly efficient, especially for short bills, yet the most important participants still exercise market power and earn substantial surplus. In practice, the paper implies that improving auction design can at best generate modest revenue gains, and that those gains are concentrated in longer-maturity securities where bid heterogeneity and shading are strongest.